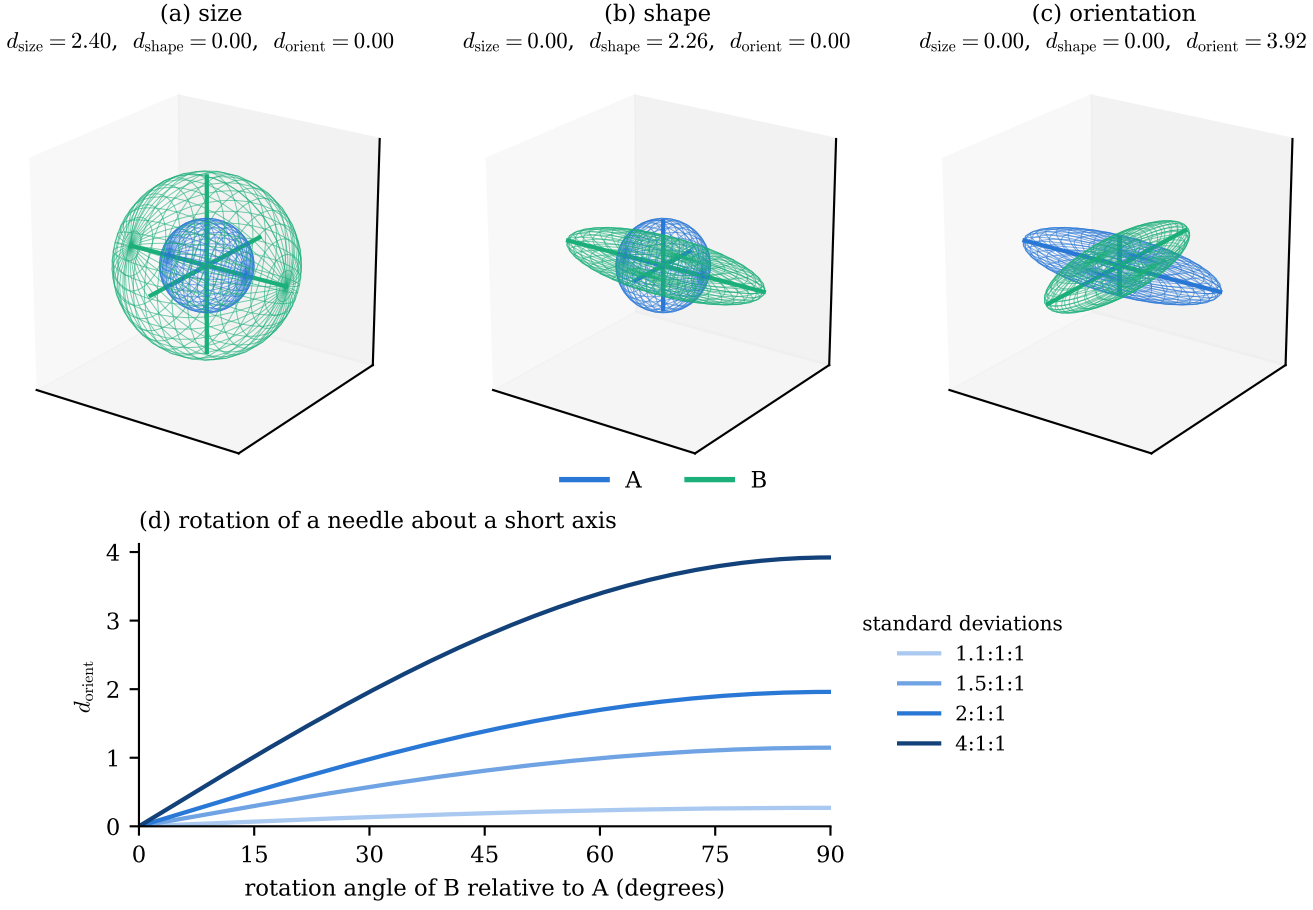

(a) size
$d_{\text{size}} = 2.40,\ d_{\text{shape}} = 0.00,\ d_{\text{orient}} = 0.00$
(b) shape
$d_{\text{size}} = 0.00,\ d_{\text{shape}} = 2.26,\ d_{\text{orient}} = 0.00$
(c) orientation
$d_{\text{size}} = 0.00,\ d_{\text{shape}} = 0.00,\ d_{\text{orient}} = 3.92$
A
B
(d) rotation of a needle about a short axis
d_{orient}
0
1
2
3
4
0
15
30
45
60
75
90
rotation angle of B relative to A (degrees)
standard deviations
1.1:1:1
1.5:1:1
2:1:1
4:1:1